6CCSACCM
Cooperation, Coordination and Multi-Agent Systems

Week 5

Cooperative games (part II)

Maria Polukarov

Understanding the solution

- A given instance may have many stable matchings.
- Would Gale-Shapley always produce the same matching, independently of the order in which men propose?
- If so, which one?

Stable matchings: $\{A - I, B - J, C - K\}$, $\{B - I, A - J, K - C\}$

Anna:	Jacob $\succ$ Isaac $\succ$ Ken	Isaac:	Anna $\succ$ Bella $\succ$ Clare
Bella:	Isaac $\succ$ Jacob $\succ$ Ken	Jacob:	Bella $\succ$ Anna $\succ$ Clare
Clare:	Isaac $\succ$ Jacob $\succ$ Ken	Ken:	Anna $\succ$ Bella $\succ$ Clare

Understanding the solution

Definition

A women x is a **valid partner** of man y if there exists a stable matching M with $x - y \in M$.

Stable matchings: $\{A - I, B - J, C - K\}$, $\{B - I, A - J, K - C\}$

Both Anna and Bella are valid partners for Isaac and Jacob.
Clare is the only valid partner for Ken.

Anna:	Jacob $\succ$ Isaac $\succ$ Ken	Isaac:	Anna $\succ$ Bella $\succ$ Clare
Bella:	Isaac $\succ$ Jacob $\succ$ Ken	Jacob:	Bella $\succ$ Anna $\succ$ Clare
Clare:	Isaac $\succ$ Jacob $\succ$ Ken	Ken:	Anna $\succ$ Bella $\succ$ Clare

Man-optimal matching

In a man-optimal matching, each man receives its best valid partner.

- Is this a perfect matching?
- Is it stable?

$$\{A - I, B - J, C - K\}, \text{ is man-optimal.}$$

Anna:	Jacob $\succ$ **Isaac** $\succ$ Ken	Isaac:	**Anna** $\succ$ Bella $\succ$ Clare
Bella:	Isaac $\succ$ **Jacob** $\succ$ Ken	Jacob:	**Bella** $\succ$ Anna $\succ$ Clare
Clare:	Isaac $\succ$ Jacob $\succ$ **Ken**	Ken:	Anna $\succ$ Bella $\succ$ **Clare**

Theorem

The DA algorithm (with men proposing) outputs the man-optimal matching.

Proof of theorem

Proof by contradiction:

- suppose some man is matched with someone who is not his best valid partner
- then some man is rejected by a valid partner
- consider the first time in the execution of the algorithm when this happens
- let I be the man rejected at this point
- let A be the women who rejects him at this point
  - she rejects I in favour of another man J

Proof of theorem

- t: time of first rejection by valid partner
- I: man rejected at t; A: woman rejecting at t
- J: the partner of A right after t
- let M be a stable matching with $A - I \in M$
- let B be the partner of J in M
- at time t, J has not been rejected by B
  - this is because B is a valid partner of J
- hence, J has not yet proposed to B
- this means that J prefers A to B
- also, A prefers J to I
- $\Rightarrow M$ is unstable! $\square$

Women pessimality

- Does man optimality come at the expense of women?
- In a women-pessimal assignment, each women gets her worst valid partner.

Theorem

The DA algorithm (with men proposing) outputs the women-pessimal matching.

Proof of theorem

- suppose under DA women A is matched to man J but J is not the worst valid partner for A
- let I be the worst valid partner for A
- A prefers J to I
- there is a stable matching M with $A - I \in M$
- let B be the partner of J in M
- since DA outputs the man-optimal matching, J prefers A to B
- $\Rightarrow M$ is unstable! $\square$

Strategyproofness

- We say that in an instance $I = \langle (\succ_x) x \in X, (\succ_y) y \in Y \rangle$ a man $y \in Y$ can benefit from misreporting his preferences under algorithm A if:
  - given I, algorithm A matches y to x;
  - there exists a ranking $\succ'$ over X such that if y reports $\succ'$ instead of $\succ_y$ (and other reports are unchanged), algorithm A matches y to x' such that $x' \succ_y x$.
- Same definition for women.
- We say that an algorithm A is **strategyproof** if on every instance no agent (man or woman) can benefit from misreporting their preferences under A.

Is Gale-Shapley strategyproof?

THANK YOU!